

Date: 28/07/26

V. Hussain

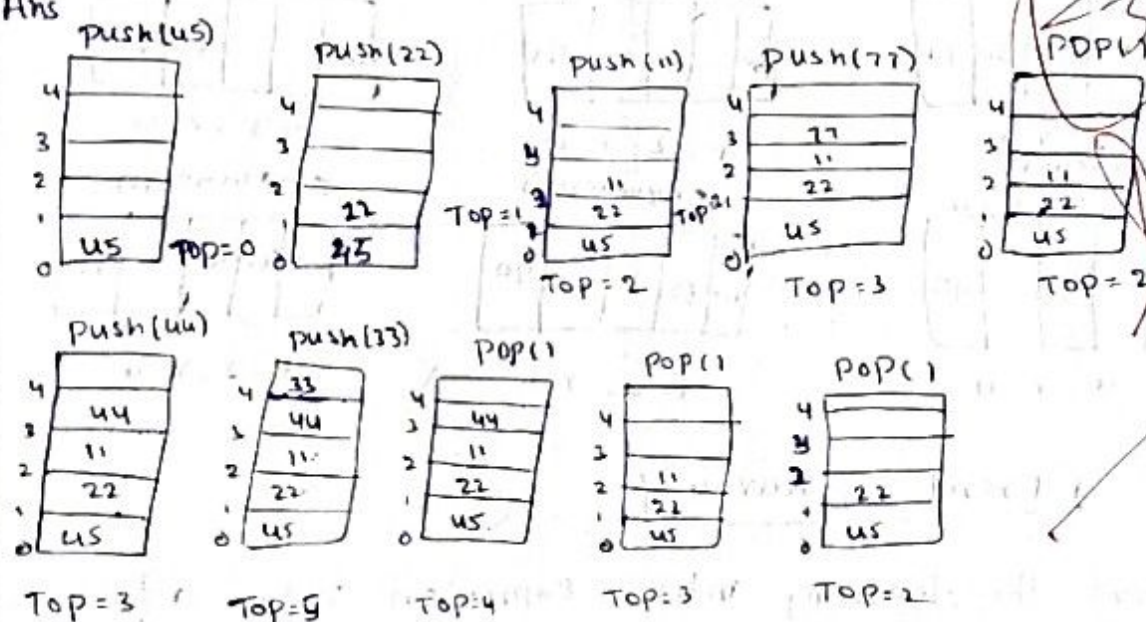
CSA0315

Reg. No: 192524179

Analytical problem solving (Test):

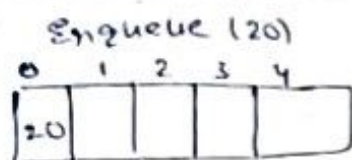
- ① Perform the following operations in LIFO data structure having an array size of 5. push(45), push(22), push(11), push(77), pop(), push(44), push(33), pop(), pop(), pop(). Print with a neat sketch Explain in detail.

Ans

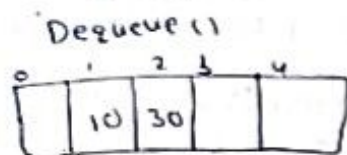


Top Position = 1
Top Element = 22

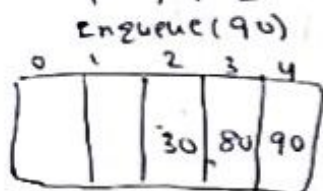
- ② Perform the following operations in circular queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90), finally display the front and rear elements with its position. Discuss in detail.



F = 0; Y = 0



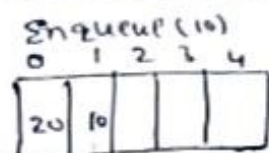
F = 1; Y = 2



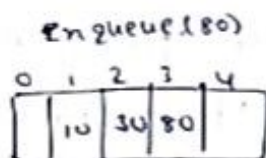
F = 2; Y = 4



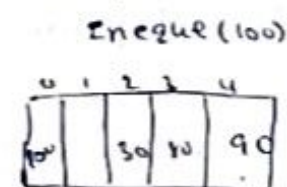
F = 4; Y = 0



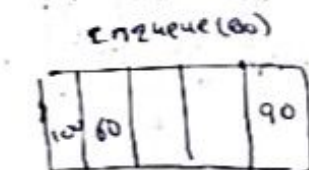
F = 0; Y = 1



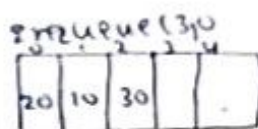
F = 1; Y = 3



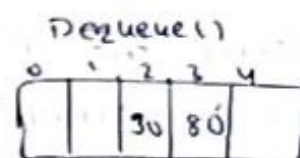
F = 2; Y = 0



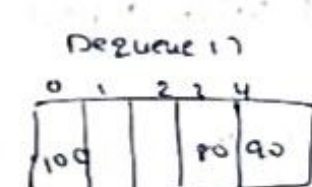
F = 4; Y = 1



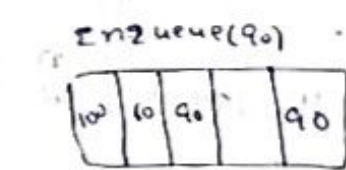
F = 0; Y = 2



F = 2; Y = 3



F = 3; Y = 0



F = 2; Y = 4

Front: 2; rear: 4

Convert the following infix expression into a postfix expression

$$B - (C / D + (E \% F * G) / H) * J.$$

also evaluation the given postfix expression. a) 9 3 8 4 / * +

b) 6 2 + 6 3 / *

Scanner postfix	stack	operation					
B	<table border="1"><tr><td></td><td></td><td></td><td></td><td></td></tr></table>						B
-	<table border="1"><tr><td>-</td><td></td><td></td><td></td><td></td></tr></table>	-					B
-							
c	<table border="1"><tr><td>-</td><td>c</td><td></td><td></td><td></td></tr></table>	-	c				Bc
-	c						
c	<table border="1"><tr><td>-</td><td>c</td><td></td><td></td><td></td></tr></table>	-	c				Bc
-	c						
/	<table border="1"><tr><td>-</td><td>c</td><td>/</td><td></td><td></td></tr></table>	-	c	/			Bc
-	c	/					
b d	<table border="1"><tr><td>-</td><td>c</td><td>/</td><td></td><td></td></tr></table>	-	c	/			Bcd
-	c	/					
+	<table border="1"><tr><td>-</td><td>c</td><td>+</td><td></td><td></td></tr></table>	-	c	+			Bcd/
-	c	+					
c	<table border="1"><tr><td>+</td><td>c</td><td>+</td><td>c</td><td></td></tr></table>	+	c	+	c		Bcd/
+	c	+	c				
E	<table border="1"><tr><td>-</td><td>c</td><td>+</td><td>c</td><td></td></tr></table>	-	c	+	c		Bcd/c
-	c	+	c				
%	<table border="1"><tr><td>-</td><td>c</td><td>+</td><td>c</td><td>%</td></tr></table>	-	c	+	c	%	Bcd/c
-	c	+	c	%			
F	<table border="1"><tr><td>-</td><td>c</td><td>+</td><td>c</td><td>%</td></tr></table>	-	c	+	c	%	Bcd/cF
-	c	+	c	%			
*	<table border="1"><tr><td>-</td><td>c</td><td>+</td><td>c</td><td>*</td></tr></table>	-	c	+	c	*	Bcd/cF%
-	c	+	c	*			
u	<table border="1"><tr><td>-</td><td>c</td><td>+</td><td>c</td><td>*</td></tr></table>	-	c	+	c	*	Bcd/cF%u
-	c	+	c	*			
/	<table border="1"><tr><td>-</td><td>c</td><td>+</td><td>c</td><td>*</td></tr></table>	-	c	+	c	*	Bcd/cF%u*
-	c	+	c	*			
H	<table border="1"><tr><td>-</td><td>c</td><td>+</td><td>c</td><td>*</td></tr></table>	-	c	+	c	*	Bcd/cF%u*x
-	c	+	c	*			
)	<table border="1"><tr><td></td><td></td><td></td><td></td><td></td></tr></table>						Bcd/cF%u*xT
x	<table border="1"><tr><td>-</td><td>x</td><td></td><td></td><td></td></tr></table>	-	x				Bcd/cF%u*xT
-	x						
J	<table border="1"><tr><td>-</td><td></td><td></td><td></td><td></td></tr></table>	-					Bcd/cF%u*xTJ
-							
over	<table border="1"><tr><td></td><td></td><td></td><td></td><td></td></tr></table>						Bcd/cF%u*xTJ+

a) 9 3 8 4 / x +

Scanning

stack

operation

9

9

3

9 3

8

9 3 8

4

9 3 8 4

/

9 3 2

8/4

8/4

x

9 6

3x2

3x2

+

15

9+6

9+6

b) 62+63 / x

Scanning

stack

operation

6

6

2

6 2

+

8

8

6

8 6

6+2

3

8 6 3

/

8 2

6/3

x

16

8x2

8x2